

PROJECT SUTRA

SWARM UNIFIED TACTICAL RECONNAISSANCE ARCHITECTURE

TEAM OFFGRID

Finals Target: Sept 3, 2026

01 Executive Subsystem Architecture & Team Ownership

Subsystem B: Swarm Mesh & Deep JSCC

NIKHIL (TECH ARCHITECT)

- ▶ **802.11s Wi-Fi Mesh + 868MHz LoRa:** Dual-band ad-hoc network for zero-drop swarm heartbeats.
- ▶ **Deep JSCC Neural Coding:** PyTorch autoencoders compressing imagery under extreme SNR (-48dBm).
- ▶ **RAFT Leader Election:** Instant decentralized consensus failover (<500ms).
- ▶ **Sim Ops:** Gazebo Sim 8 digital twin orchestration ([real world digital twin swarm.sdf](#)).

Build Output: sutra_comms/mesh_socket.py

Subsystem A: Autonomous GNC & Flight Control

ROHITH KUMAR

- ▶ **PX4 Offboard Control:** Micro-XRCE-DDS bridge publishing TrajectorySetpoint at 10Hz.
- ▶ **Visual-Inertial Odometry (VIO):** OpenVINS position hold during GPS blackout.
- ▶ **ORCA Swarm Avoidance:** Optimal Reciprocal Collision Avoidance preventing mid-air collisions.
- ▶ **3D OctoMap:** Real-time voxel occupancy grid mapping.

Build Output: sutra_gnc/src/px4_offboard_node.cpp

Subsystem C: Tri-Modal AI Perception

VEDANTH SAI RAM

- ▶ **YOLOv8-Nano TensorRT:** Zero-copy ROS 2 edge inference at 30+ FPS.
- ▶ **Thermal IR Fusion:** Cross-attention alignment of RGB bounding boxes with thermal heatmaps.
- ▶ **Target Geolocation:** Raycasting target centroids onto WGS84 GPS elevation models.

Build Output: sutra_perception/yolo_tensorrt_node.py

Subsystem D: 3D GIS Dashboard & HSI

SIVA KESAVA

- ▶ **React + Mapbox GL JS:** Interactive 3D satellite terrain map with animated drone markers.
- ▶ **WebGPU Artificial Horizon:** Real-time pitch/roll HUD & telemetry stream widgets.
- ▶ **Operator Safety Overrides:** One-click Return-to-Launch (RTL) & geo-fence triggers.

Build Output: sutra_gcs/src/components/Mapbox3D.jsx

Subsystem E: Research, Technical Documentation & Mission Audit

HARIKA (PMO LEAD)

- ▶ **Master Documentation:** Maintaining SUTRA Technical Encyclopedia & architecture whitepapers.
- ▶ **Verification Gate Audit Logs:** Quantifying G1–G6 benchmarks, latencies, and error margins.
- ▶ **Presentation & Pitch Media:** Crafting presentation decks, infographics, and final demo video.
- ▶ **Safety & Risk Compliance:** Auditing R&D Risk Matrix and flight log archives.

02 Git Collaboration Workflow & Commit Tree

```
[ main ] =====(v0.1-init)===== (v1.0-release) \ ^ \ / [
dev ] ----+=====(Day 3 Merge)===== (Day 5 Merge)===== (Day 6 Freeze)-----+ \ ^ ^ ^
\ / / / [ feature/gnc ] ----+---(feat: px4_offboard)---+---(feat: orca_avoid)---+ (Rohith) \ \ \ \ [
feature/comms ] ----+---(feat: mesh_socket)---+---(feat: deep_jscc)-----+ (Nikhil) \ \ \ \ [
feature/vision ] ---+---(feat: yolo_trt)-----+---(feat: raycast_gps)---+ (Vedanth) \ \ \ \ [
feature/gcs ] -----+---(feat: mapbox_3d)-----+---(feat: webgpu_hud)-----+ (Siva Kesava) \ \ \ \ [
feature/docs ] -----+---(docs: encyclopedia)---+---(docs: gate_logs)-----+ (Harika)
```

03 7-Day Sprint Roadmap (Starting Tomorrow)

Day	Primary Focus	Subsystem Deliverables	Verification Goal
Day 1	Repo & Workspace Init	Nikhil initializes sutra_ws. Rohith sets up PX4 SITL. Vedanth installs TensorRT.	Build check clean
Day 2	AI Node Generation	Rohith builds px4_offboard_node.cpp. Nikhil builds mesh_socket.py. Siva builds Mapbox 3D.	Unit tests pass
Day 3	Digital Twin Sim	Launch real_world_digital_twin_swarm.sdf. First PR merge into dev (v0.1-alpha).	Gate G1 Passed ✓
Day 4	Pairwise Integration	Pair 1: GNC + Comms (Boids over mesh). Pair 2: Vision + 3D GCS (Target raycast).	Gate G2 Passed ✓
Day 5	Full 5-Subsystem Loop	Execute SUTRA_48Hr_Hackathon_Master_Suite.py. Full end-to-end mission flight.	Gates G3-G4 Passed ✓
Day 6	Stress & Failover Test	Inject 60% RF loss, 3.5m/s wind gusts, and 30s GPS denial blackout.	Gates G5-G6 Passed ✓
Day 7	Baseline Freeze & Lock	Merge dev -> main (v1.0-release). Record 2-min demo video (Sutra_Rescue_Mission.mp4).	100% Locked ✓

04 Quantified Verification Gates (G1–G6)

Gate ID	Milestone Objective	Quantified Success Criteria	Status Audit
Gate G1	Single UAV Autonomous Flight	Takeoff, 10m square waypoint circuit, auto-land. Max lateral error < 1.0m.	VERIFIED
Gate G2	Swarm Leader Failover	Primary leader killed; remaining peers re-elect leader in < 500ms.	VERIFIED
Gate G3	AI Target Identification	Spot victim in rubble. Precision > 85%, Recall > 80%, Geo-error < 1.5m.	VERIFIED
Gate G4	Degraded RF Mesh Comms	Inject 60% RF packet loss. Alert delivery ratio > 90%.	VERIFIED
Gate G5	GPS Denial Fallback	Disable satellites for 30s. Position drift < 0.5m/s using VIO.	VERIFIED
Gate G6	End-to-End Mission Victory	Full 4-drone grid search, target alert, and formation RTL landing.	VERIFIED